Simulation Visualization Dashboard

Overview

Panisim provides a web-based visualization dashboard to monitor vessel simulation in real-time. The dashboard offers a comprehensive interface for monitoring vessel state, visualizing trajectories, viewing sensor data, and interacting with the simulation.

The visualization system consists of two main components:

1. **2D Dashboard** - A data-rich interface showing vessel state, charts, and control inputs
2. **3D Visualization** - A three-dimensional view of the vessel in its environment with position and orientation

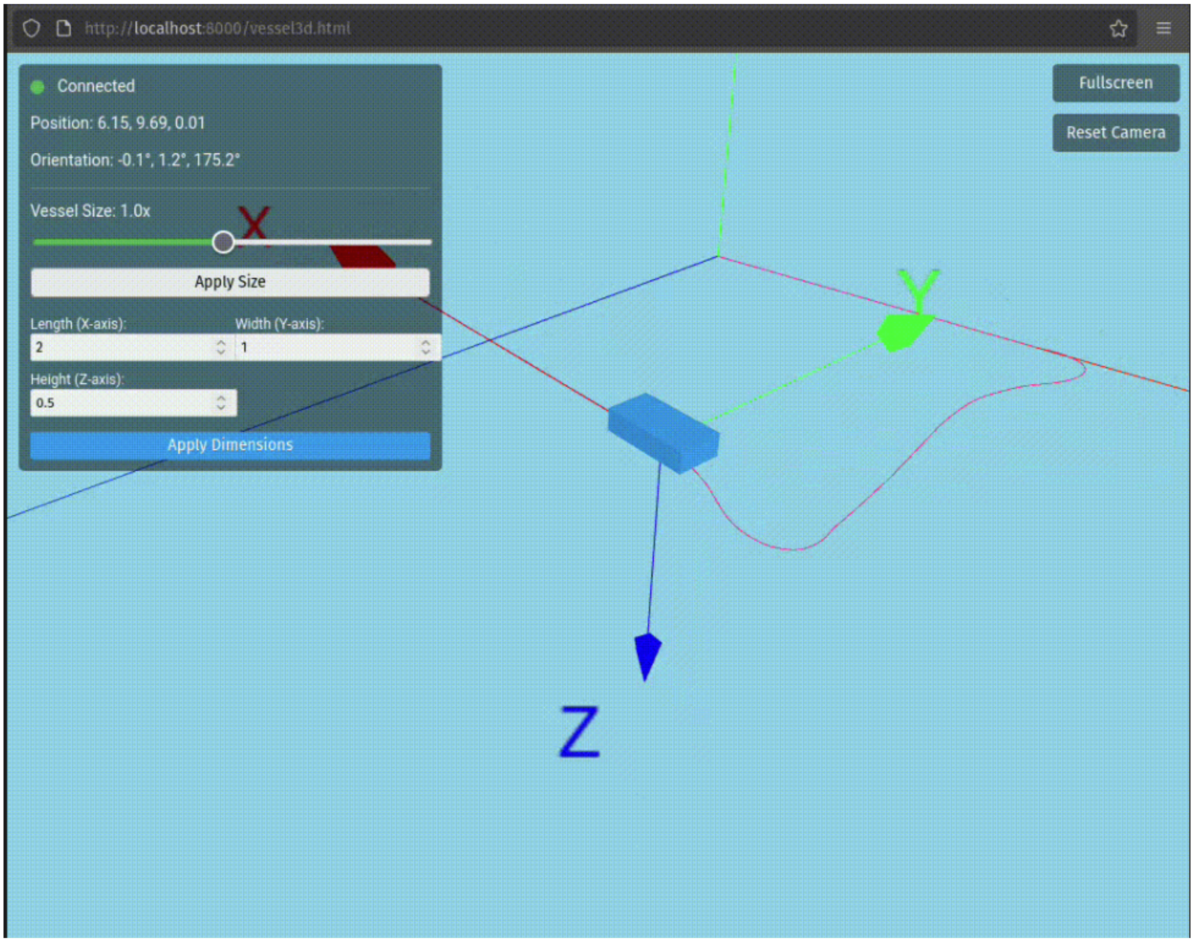
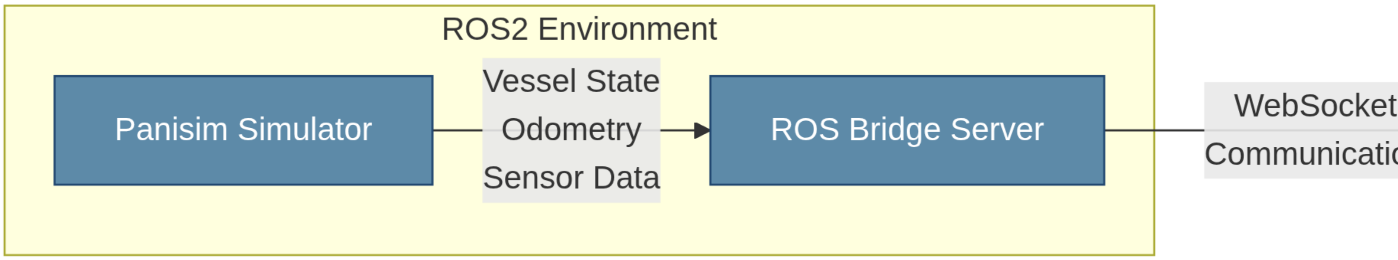


Figure 1: Way-point trackingSimulation Visualization



Starting the Visualization

The web interface is automatically started when launching the Panisim simulator. It uses:

1. A lightweight HTTP server to host the web files
2. The ROS Bridge server for WebSocket communication with ROS2

The visualization launches when running the simulation:

```
ros2 launch gnc gnc_launch.py
```

Once started, you can access the dashboard by opening your web browser:

- **Main Dashboard**: `http://localhost:8000/`
- **3D Visualization**: `http://localhost:8000/vessel3d.html`

Main Dashboard

The main dashboard (index.html) provides a comprehensive overview of the vessel's state, sensor data, and control inputs.

Key Features

1. **Vessel Selection**: Switch between different vessels in the simulation
2. **Connection Status**: Real-time display of connectivity to the ROS2 environment
3. **Vessel Information**: Display of key vessel parameters (position, velocity, orientation)
4. **Data Visualizations**: Real-time charts for:

   - Vessel path and trajectory
   - Velocities (linear and angular)
   - Orientation (roll, pitch, yaw)
   - Control surface positions
   - Thruster states

Dashboard Layout

The dashboard is organized in a responsive grid layout with cards for different data categories:

Section	Description
Vessel Info	Basic vessel parameters including position, velocities, and orientation
Path Visualization	2D trajectory visualization showing vessel movement in X-Y plane
Vessel Velocities	Charts for surge (u), sway (v), and heave (w) velocities
Vessel Angular Rates	Charts for roll rate (p), pitch rate (q), and yaw rate (r)
Orientation	Charts for roll ( ), pitch ( ), and yaw ( ) angles
Control Surfaces	Visual representation of control surface positions
Thrusters	Visual representation of thruster RPMs

Real-time Data Updates

The dashboard updates in real-time with configurable update rates. Data collection begins automatically when connecting to a vessel and includes:

- Subscribe to `/[vessel_name]/odometry_sim` for position and velocity data
- Subscribe to `/[vessel_name]/vessel_state` for complete vessel state including actuators
- Optional subscription to additional sensor topics

3D Visualization

The 3D visualization (vessel3d.html) provides an interactive three-dimensional view of the vessel in its environment.

Key Features

1. **Real-time Movement**: The vessel model moves and rotates according to simulation data
2. **Trajectory Path**: A colored trail showing the vessel's path through the environment
3. **Customizable View**: Camera controls for rotating, panning, and zooming
4. **Vessel Dimensions**: Adjustable vessel size and proportions to match different vessel types
5. **Full-screen Mode**: Expand the visualization to utilize the entire screen

Controls

The 3D view offers several interactive controls:

- **Orbit**: Click and drag to rotate the camera around the vessel
- **Pan**: Right-click and drag (or middle-click and drag) to move the camera
- **Zoom**: Scroll wheel to zoom in and out
- **Reset Camera**: Button to restore the default camera position
- **Fullscreen**: Button to toggle fullscreen mode

Vessel Model Customization

You can adjust the vessel's appearance through the control panel:

- **Vessel Size**: Overall scaling of the vessel model
- **Dimensions**: Individual control over length, width, and height proportions
- **Apply Changes**: Button to update the model with the new dimensions

Integration with ROS2

The visualization dashboard communicates with ROS2 using the ROS Bridge Server, which provides WebSocket connectivity between the ROS2 environment and web clients.

Communication Flow

1. **ROS2 Topics to WebSockets**: The ROS Bridge Server converts ROS2 topic messages to WebSocket messages
2. **Web Client Processing**: JavaScript code processes the WebSocket messages and updates the visualizations
3. **Bidirectional Communication**: The system supports both subscribing to and publishing ROS2 topics

Required ROS2 Components

To enable the visualization, the launch file includes:

```
# HTTP server for web files
ExecuteProcess(
    cmd=['python3', '-m', 'http.server', '8000', '--directory', web_build_dir],
    name='http_server',
    output='screen'
),

# ROS Bridge server for WebSocket communication
IncludeLaunchDescription(
    AnyLaunchDescriptionSource(rosbridge_launch)
),
```

Topic Subscriptions

The visualization automatically discovers vessel topics by querying available ROS2 topics. It looks for the following patterns:

- `/<vessel_name>/odometry_sim`: For vessel position and velocity
- `/<vessel_name>/vessel_state`: For complete vessel state including actuators

Implementation Details

The visualization dashboard is built using modern web technologies:

- **Chart.js**: For responsive, interactive data visualization
- **three.js**: For 3D rendering of the vessel
- **roslibjs**: For ROS2 communication via WebSockets
- **Responsive Design**: Automatically adapts to different screen sizes

Key JavaScript Components

The system is organized into the following components:

1. **VesselManager**: Handles vessel discovery, selection, and configuration
2. **VesselVisualizer**: Manages data collection, processing, and visualization
3. **3D Rendering**: Handles the three-dimensional visualization of vessels

Vessel State Representation

The dashboard visualizes the complete vessel state model:

- **Position**: x, y, z (in meters)
- **Linear Velocity**: u, v, w (in m/s)
- **Angular Velocity**: p, q, r (in rad/s)
- **Orientation**: , , (in degrees)
- **Actuators**: Control surface angles (in degrees) and thruster RPMs

Customization Options

The visualization can be customized in several ways:

1. **Update Rates**: Chart refresh rates can be adjusted
2. **Data Window**: The amount of historical data displayed in charts
3. **Vessel Configuration**: The number of control surfaces and thrusters displayed
4. **3D Model**: Vessel dimensions and appearance
5. **Chart Scaling**: Auto-rescaling and manual zoom/pan on charts

Troubleshooting

Connection Issues

If the dashboard shows "Disconnected" status:

1. Ensure the ROS Bridge server is running
2. Check that WebSocket port 9090 is accessible
3. Verify the simulator is publishing required topics

Missing Vessel Data

If vessel data isn't displaying:

1. Select a vessel from the dropdown menu
2. Check ROS2 topics are being published:

   ```
   ros2 topic list | grep vessel
   ```

3. Verify message publication with:

   ```
   ros2 topic echo /<vessel_name>/odometry_sim
   ```

Browser Compatibility

The visualization works best with modern browsers:

- Google Chrome
- Mozilla Firefox

- Microsoft Edge
- Brave Browser

Conclusion

The Panisim visualization dashboard provides a comprehensive interface for monitoring and interacting with simulated vessels. With its real-time updates, interactive charts, and 3D visualization, it offers a powerful tool for understanding vessel behavior and validating custom control or guidance algorithms.